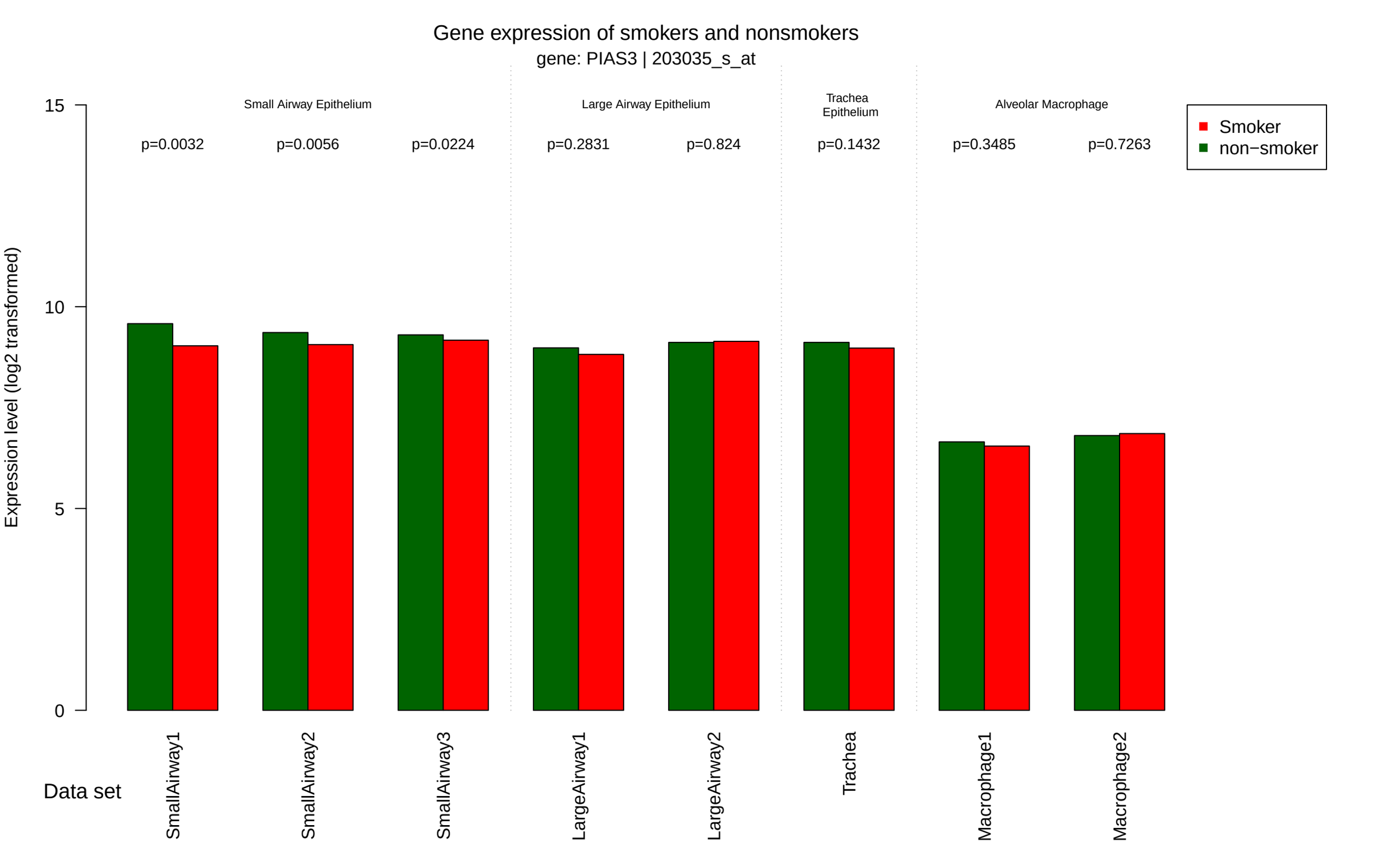
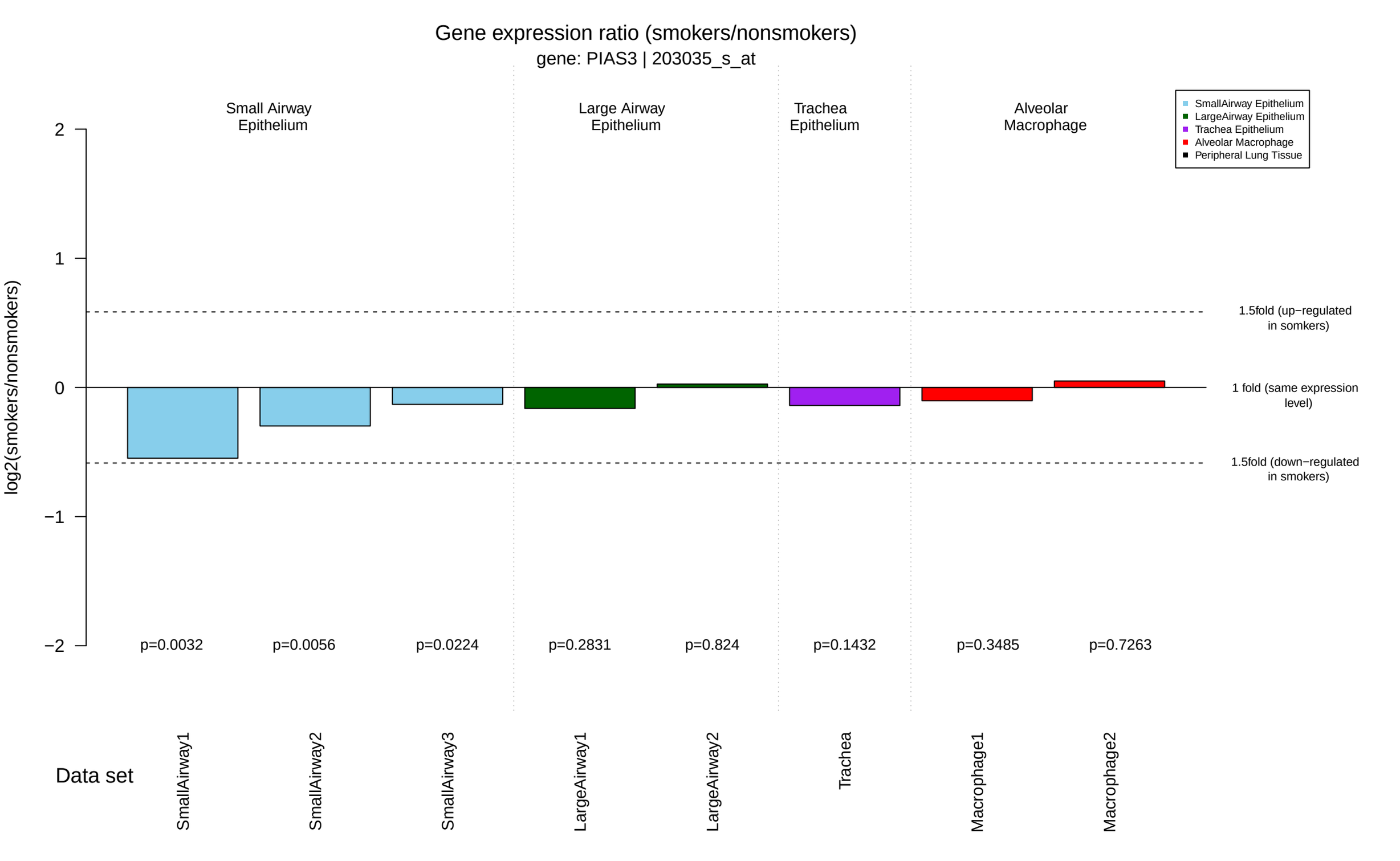
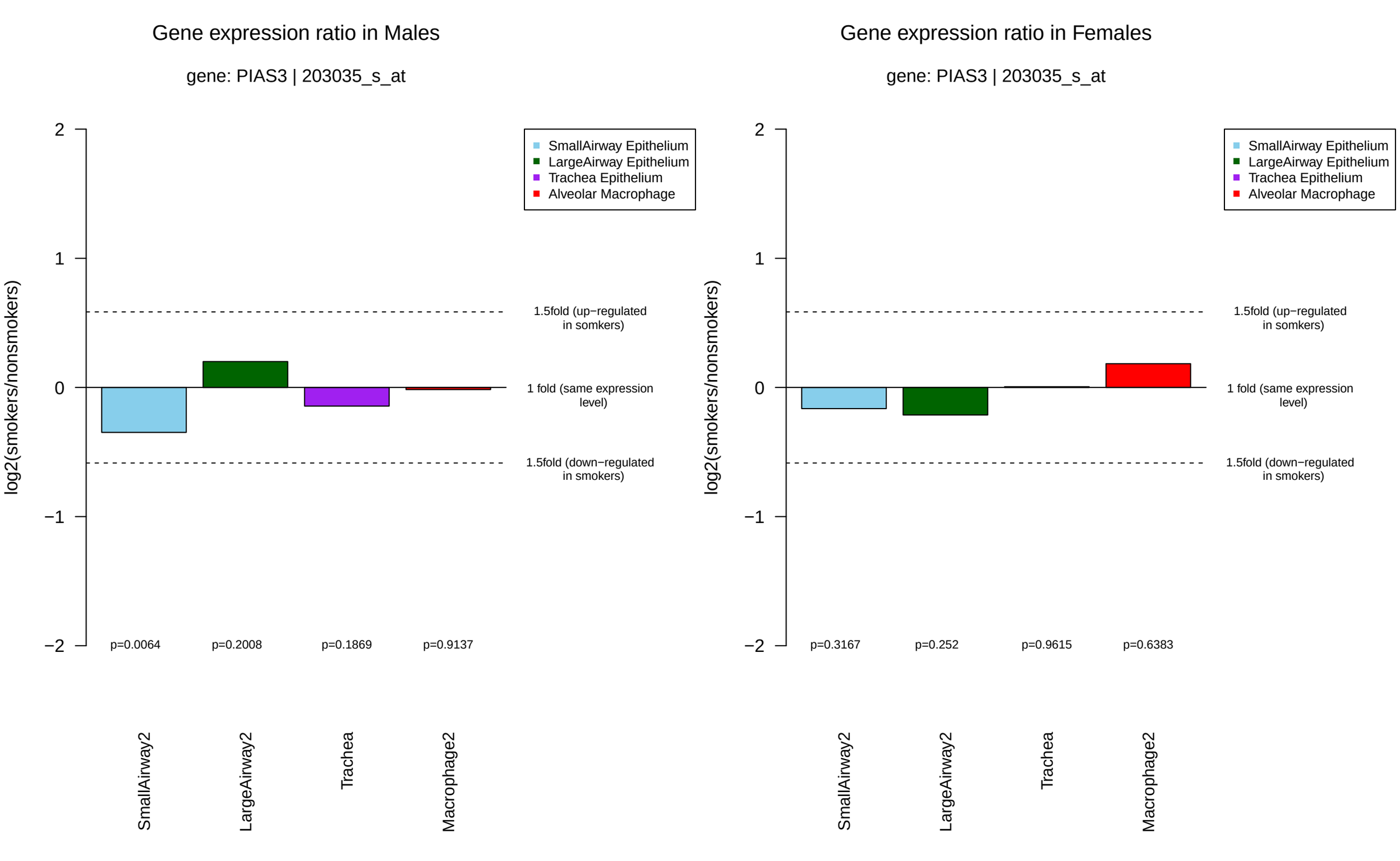
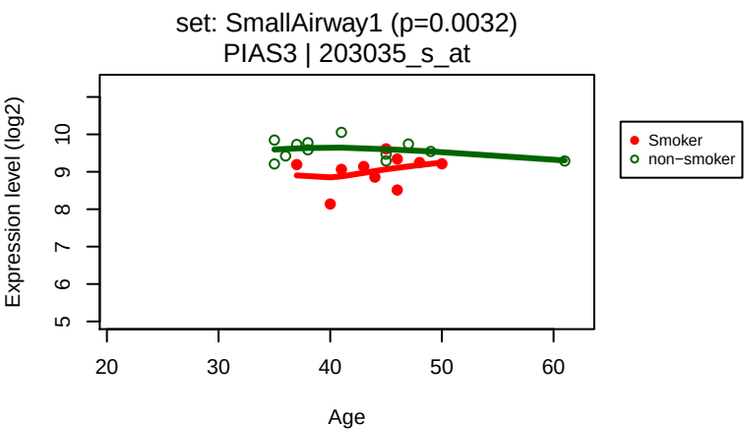
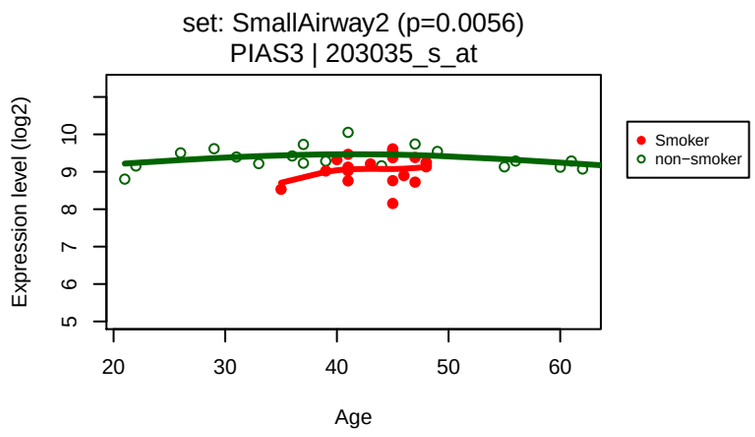
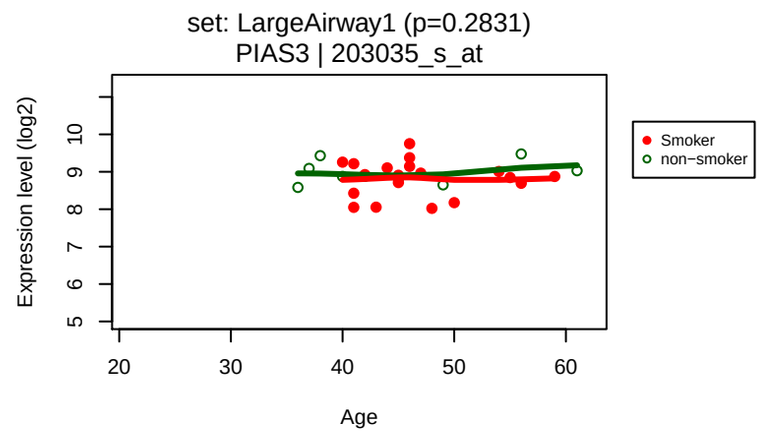

Gene expression of smokers and nonsmokers
gene: PIAS3 | 203035_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0032
p=0.0056
p=0.0224
p=0.2831
p=0.824
p=0.1432
p=0.3485
p=0.7263
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

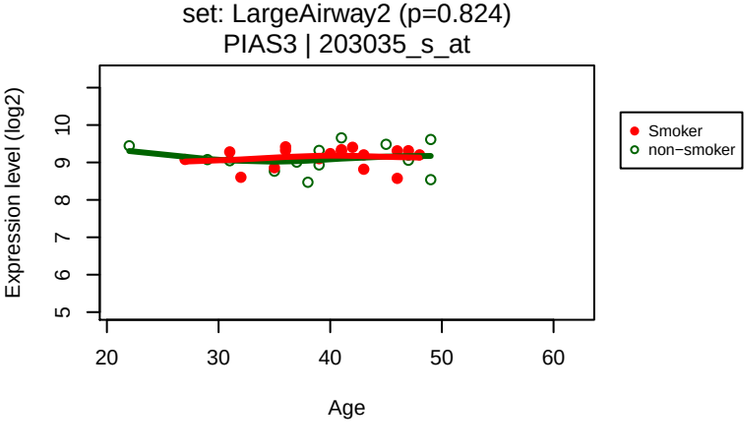

Gene expression ratio (smokers/nonsmokers)
gene: PIAS3 | 203035_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0032
p=0.0056
p=0.0224
p=0.2831
p=0.824
p=0.1432
p=0.3485
p=0.7263
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

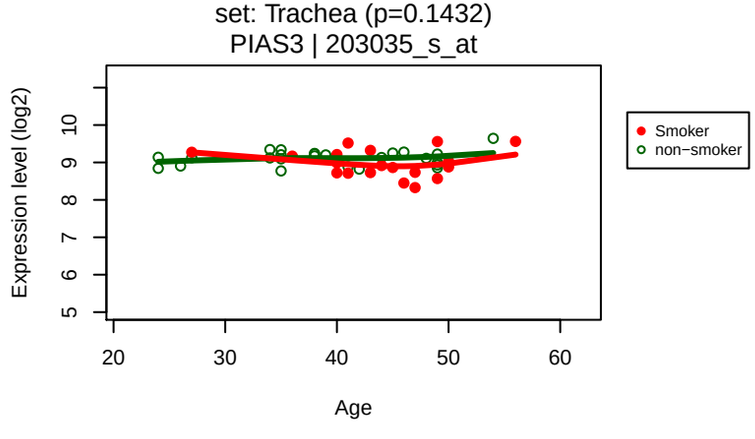

Gene expression ratio in Males
gene: PIAS3 | 203035_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0064
p=0.2008
p=0.1869
p=0.9137
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIAS3 | 203035_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3167
p=0.252
p=0.9615
p=0.6383
SmallAirway2
LargeAirway2
Trachea
Macrophage2

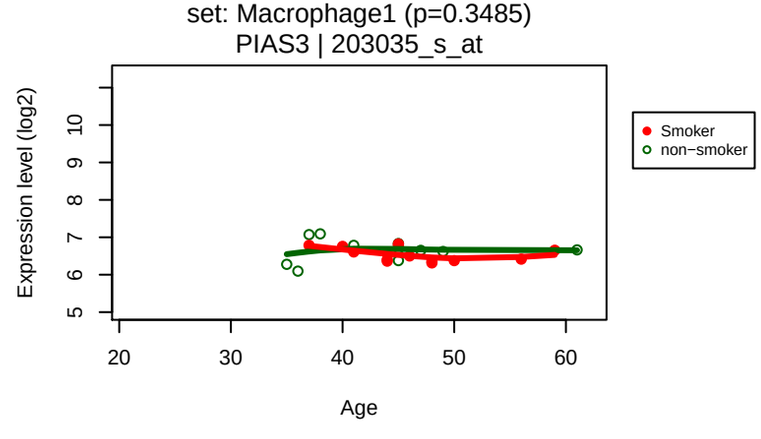

set: SmallAirway1 (p=0.0032)
PIAS3 | 203035_s_at
Expression level (log2)
Age
Smoker
non−smoker

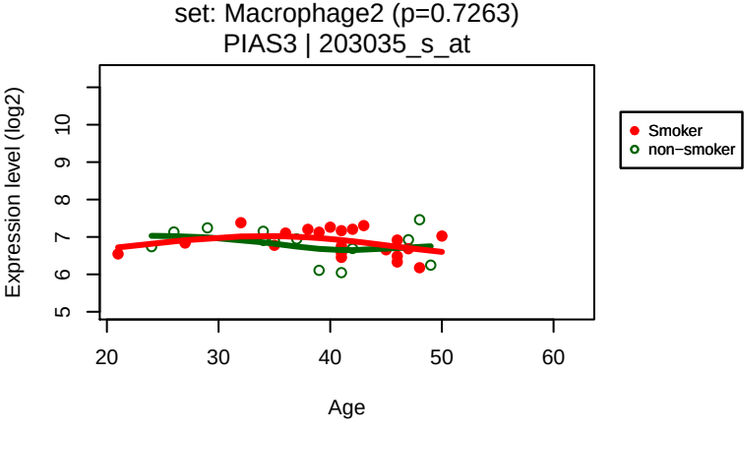

set: SmallAirway2 (p=0.0056)
PIAS3 | 203035_s_at
Expression level (log2)
Age
Smoker
non−smoker

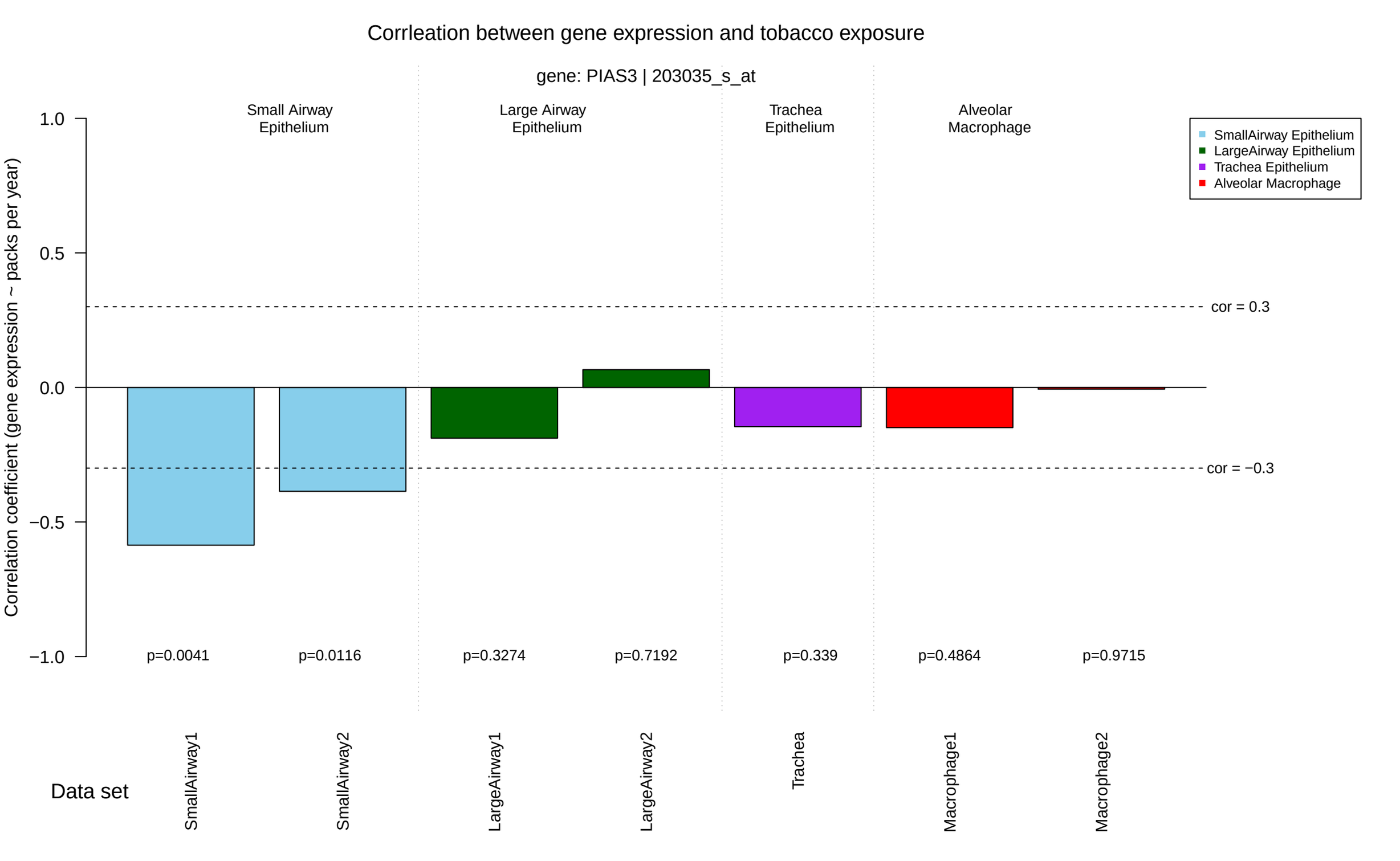

set: LargeAirway1 (p=0.2831)
PIAS3 | 203035_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.824)
PIAS3 | 203035_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1432)
PIAS3 | 203035_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3485)
PIAS3 | 203035_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7263)
PIAS3 | 203035_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIAS3 | 203035_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0041
p=0.0116
p=0.3274
p=0.7192
p=0.339
p=0.4864
p=0.9715
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2